



When the Sky Becomes a Crisis

Meteorological Extremes in France — 35 Years of Data

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4DVST — Data Visualisation · June 2026

Météo-France Open Data

Metabase

PostgreSQL

Docker

Agenda

01 — Context & Data

The challenge, our source, what a crisis means

02 — Technical Approach

Stack, pipeline, and how data flows to the dashboard

03 — Dashboard Walkthrough

5 interactive visualisations and what they reveal

04 — Key Findings

The 3 forces that shape France's storm geography

The Story We're Telling

15 million

weather observations · 28 French departments · 1990 to 2025

"Climate change is often described as a distant threat. Our goal: make it visible, measurable — and undeniable."



Where?

Which departments are most exposed to extreme rainfall?



When?

How have crisis events evolved over 35 years?



Why?

What geography and climate forces shape exposure?

Why This Matters



Flooding

Infrastructure damage, road closures, property losses across entire regions



Public Safety

Evacuations, emergency response mobilised, lives placed at risk



Economic Cost

Billions in annual losses across agriculture, transport and industry

*Understanding which territories are most exposed is the **first step to anticipating what climate change will intensify** in the coming decades.*

Our Data Source

Météo-France Open Data

France's national meteorological service — one of the world's most reliable public weather datasets. Freely accessible via meteo.data.gouv.fr, updated continuously with validated observations.

What we collected per station

- Daily rainfall in millimetres (RR)
- Min / Max / Mean temperature
- Wind speed and peak gusts
- Station location: lat, lon, altitude
- Data quality codes (validated data only)

28

representative departments

35

years of daily observations

15M+

rows imported and processed

What Is a "Meteorological Crisis"?

*No universal threshold exists. We chose a **data-driven definition** anchored in Météo-France standards — strict enough to be meaningful, broad enough to capture real risk.*



≥ 20 mm

of rainfall in a **single day**

Météo-France official "heavy rain" indicator



≥ 2 days

consecutive days above that threshold

Sustained saturation = real flooding risk

Detection method: **Gaps & Islands** — identifies uninterrupted sequences of rainy days, grouped into episodes with start date, end date, total rainfall and department

Technical Stack



Docker

Containerised ·
Reproducible on any
machine



PostgreSQL

Data storage & crisis
detection engine



Python

API download & CSV
import pipeline



Metabase

Interactive dashboard ·
No-code visualisation



Single command deployment · All services start together · No manual configuration needed

From Raw Data to Insights — The Journey



Download

CSV files fetched from
Météo-France API · 28
departments



Store

Bulk import into
PostgreSQL · Quality
filter (validated data
only)



Detect

Crisis episodes
identified · Materialised
views for instant
queries



Visualise

5 interactive charts in
Metabase · Filters by
year, department

✓ **Data quality** — only observations validated by
Météo-France climatologists

✓ **Instant queries** — pre-computed crisis data,
Metabase loads in seconds

Dashboard — 5 Views, One Story

35 years · 28 departments · one dashboard



Crises per year



Crisis count map



Duration map



Altitude scatter



Top 10 departments

Filters: year · department · duration

Finding 1 — Where Storms Strike

Some regions are hit far harder than others.

The three hot zones:

- **Alpine foothills** — Haute-Savoie, Savoie, Isère
- **Mediterranean coast** — Alpes-Maritimes, Gard, Hérault
- **Atlantic Pyrenees** — Pyrénées-Atlantiques

*The **Cévenol effect**: warm, humid air rises sharply against mountain slopes — releasing enormous rainfall within hours.*

Finding 2 — 35 Years of Fluctuation ⌚

Total cumulative crisis-days per year across all 28 departments:

▲ Peak: mid-1990s

Over **6,000** cumulative crisis-days in a single year — the most intense decade on record



2024: near-historic low

~2,000 crisis-days — a significant long-term decline

*Year-to-year variability is very high. Climate signals only emerge over **decades**, not single years.*

◆ No simple upward trend — the climate is complex ·
The pattern matters more than any single year

The Sponge Effect 🧽

The map shows average crisis **duration** — a completely different pattern from crisis **count**:

⚡ Mediterranean South

Many crises · **Short** (~2 days) · Dry soils recover quickly between events

☁ Atlantic North & Brittany

Fewer crises · **Longer** (2.3+ days) · Saturated soils extend every event



"Like a saturated sponge — northern soils can't absorb more water. Every new rain event simply extends the crisis further."

The Cereal Bowl Effect 🍲

Altitude shapes exposure in a precise pattern.

🍲 Just like a cereal bowl...

Water pools at the **bottom** (low valleys) and at the **edges** (mountain foothills). The foothill zone is doubly exposed — rain falls heavily **and** flows down from the peaks above.



High altitude

More precipitation ·
Snowmelt adds to runoff



Low valleys

Water concentration · Slow
natural drainage

The 5 Most Exposed Departments

- | | | |
|----------|---|---------------------|
| 1 | Pyrénées-Atlantiques
Atlantic exposure + Pyrenean orographic lift | 5,598 crises |
| 2 | Haute-Savoie
Alpine foothills — the double bowl effect | 5,535 crises |
| 3 | Savoie
Concentrated valley rainfall + snowmelt convergence | 5,346 crises |
| 4 | Alpes-Maritimes
Mediterranean Cévenol episodes — intense and fast | 4,877 crises |
| 5 | Isère
Alpine drainage convergence — runoff from multiple valleys | 4,871 crises |

3 Forces That Shape France's Storm Geography



Climate

The Mediterranean south generates intense, short-lived Cévenol events. The Atlantic north produces persistent, long-duration rains that saturate entire regions.



Soil Saturation

Already-wet northern soils cannot absorb more water. Like a full sponge — each new rainstorm simply prolongs and deepens the ongoing crisis.



Relief

Mountains force air upward, triggering heavy rainfall. Valleys concentrate runoff from above. The foothill zone faces both simultaneously — the worst of both worlds.

"It's not the volume of rain alone that creates a crisis — it's its concentration in time, space, and a landscape that has no room left."

What We Built — & How We Worked

✓ Deliverables

- Reproducible containerised pipeline (Docker)
- 15 million+ rows imported from Météo-France API & CSV
- Crisis detection with instant results (materialised views)
- 5-visualisation interactive Metabase dashboard
- Open GitHub repository, fully documented

⚡ Key Challenges

- Handling 15M rows at scale — bulk COPY + views
- Data quality: filtering out unvalidated observations
- Defining "crisis" scientifically, not just technically

🔄 Agile Process — 3 Sprints

Sprint 1 — Data exploration, stack setup, crisis definition

Sprint 2 — Pipeline, import, detection logic

Sprint 3 — Dashboard, storytelling, presentation

📌 **Collaboration via GitHub — shared repository & commit history as our versioning backbone**

Conclusion

*"Climate change is not a future risk.
It is already written in 35 years of French
weather data."*



Visible

Made the invisible pattern of extreme rain geographically tangible



Measurable

Quantified what a meteorological crisis means with a rigorous definition



Actionable

Identified the territories to prioritise for climate adaptation planning

Thank You — Questions?

We're happy to go deeper on methodology, technical choices, or any of the findings.

GitHub

[github.com/TomJous
/4DVST-project](https://github.com/TomJous/4DVST-project)

Data source

meteo.data.gouv.fr

Dashboard

**Live demo —
Metabase**

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